KAIZEN MATHS

Lesson starter

Algebraic Fractions

Solving Equations

Name _____

S1

Solve:

$\frac{x}{(x-5)} = 2$

S2

Solve:

$\frac{3}{(x+4)} = \frac{-6}{(x-5)}$

S3

Solve:

$\frac{x}{(x-3)} = 2$

S4

Solve:

$\frac{2}{(x+5)} = \frac{1}{(x+4)}$

Show your working

KAIZEN MATHS

Lesson starter

Algebraic Fractions

Solving Equations

Name _____

S1

Solve:

$\frac{x}{(x-5)} = 2$

S2

Solve:

$\frac{3}{(x+4)} = \frac{-6}{(x-5)}$

S3

Solve:

$\frac{x}{(x-3)} = 2$

S4

Solve:

$\frac{2}{(x+5)} = \frac{1}{(x+4)}$

ALG-49C4-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

Algebraic Fractions

Solving Equations

Name _____

E1

Solve:

$\frac{x}{(x-3)} = 4$

E2

Solve:

$\frac{4}{(x-1)} = \frac{-1}{(x+4)}$

E3

Solve:

$\frac{x}{(x+4)} = 2$

Show your working

KAIZEN MATHS

Exit ticket

Algebraic Fractions

Solving Equations

Name _____

E1

Solve:

$\frac{x}{(x-3)} = 4$

E2

Solve:

$\frac{4}{(x-1)} = \frac{-1}{(x+4)}$

E3

Solve:

$\frac{x}{(x+4)} = 2$

ALG-49C4-L4 · E

Show your working

KAIZEN MATHS

Exit ticket

Algebraic Fractions

Solving Equations

Name _____

E1

Solve:

$\frac{x}{(x-3)} = 4$

E2

Solve:

$\frac{4}{(x-1)} = \frac{-1}{(x+4)}$

E3

Solve:

$\frac{x}{(x+4)} = 2$

ALG-49C4-L4 · E

Show your working

KAIZEN MATHS

Exit ticket

Algebraic Fractions

Solving Equations

Name _____

E1

Solve:

$\frac{x}{(x-3)} = 4$

E2

Solve:

$\frac{4}{(x-1)} = \frac{-1}{(x+4)}$

E3

Solve:

$\frac{x}{(x+4)} = 2$

ALG-49C4-L4 · E

Show your working

Teacher answers & working

Algebraic Fractions · Solving Equations

ALG-49C4-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Solve:

$\frac{x}{(x-5)} = 2$

Answer:

$x = 10$

Working:

Step 1: State the restriction. The denominator cannot be zero, so x is not equal to 5.

Step 2: Multiply both sides by (x - 5) to clear the denominator.

$x = 2(x - 5)$

Step 3: Expand the right hand side.

$x = 2x - 10$

Step 4: Collect the x terms on one side.

$-x = -10$

Step 5: Divide by -1 to isolate x.

$x = 10.$

Check: x = 10 does not make the denominator zero.

S2

Solve:

$\frac{3}{(x+4)} = \frac{-6}{(x-5)}$

Answer:

$x = -1$

Working:

Step 1: State the excluded values. x cannot be -4 or 5.

Step 2: Multiply both sides by the common denominator (x + 4)(x - 5).

$3(x - 5) = -6(x + 4)$

Step 3: Expand both sides.

$3x - 15 = -6x - 24$

Step 4: Collect the x terms and constants.

$9x = -9$

Step 5: Divide by 9 to isolate x.

$x = -1.$

Check: x = -1 is not an excluded value.

Teacher answers & working

Algebraic Fractions · Solving Equations

ALG-49C4-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

Solve:

$\frac{x}{(x-3)} = 2$

Answer:

$x = 6$

Working:

Step 1: State the restriction. The denominator cannot be zero, so x is not equal to 3.

Step 2: Multiply both sides by (x - 3) to clear the denominator.

$x = 2(x - 3)$

Step 3: Expand the right hand side.

$x = 2x - 6$

Step 4: Collect the x terms on one side.

$-x = -6$

Step 5: Divide by -1 to isolate x.

$x = 6.$

Check: x = 6 does not make the denominator zero.

S4

Solve:

$\frac{2}{(x+5)} = \frac{1}{(x+4)}$

Answer:

$x = -3$

Working:

Step 1: State the excluded values. x cannot be -5 or -4.

Step 2: Multiply both sides by the common denominator (x + 5)(x + 4).

$2(x + 4) = 1(x + 5)$

Step 3: Expand both sides.

$2x + 8 = x + 5$

Step 4: Collect the x terms and constants.

$x = -3$

Step 5: Divide by 1 to isolate x.

$x = -3.$

Check: x = -3 is not an excluded value.

Teacher answers & working

Algebraic Fractions · Solving Equations

ALG-49C4-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

Solve:

$\frac{x}{(x-3)} = 4$

Answer:

$x = 4$

Working:

Step 1: State the restriction. The denominator cannot be zero, so x is not equal to 3.

Step 2: Multiply both sides by (x - 3) to clear the denominator.

$x = 4(x - 3)$

Step 3: Expand the right hand side.

$x = 4x - 12$

Step 4: Collect the x terms on one side.

$-3x = -12$

Step 5: Divide by -3 to isolate x.

$x = 4.$

Check: x = 4 does not make the denominator zero.

E2

Solve:

$\frac{4}{(x-1)} = \frac{-1}{(x+4)}$

Answer:

$x = -3$

Working:

Step 1: State the excluded values. x cannot be 1 or -4.

Step 2: Multiply both sides by the common denominator (x - 1)(x + 4).

$4(x + 4) = -1(x - 1)$

Step 3: Expand both sides.

$4x + 16 = -x + 1$

Step 4: Collect the x terms and constants.

$5x = -15$

Step 5: Divide by 5 to isolate x.

$x = -3.$

Check: x = -3 is not an excluded value.

Teacher answers & working

Algebraic Fractions · Solving Equations

ALG-49C4-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

Solve:

$\frac{x}{(x+4)} = 2$

Answer:

$x = -8$

Working:

Step 1: State the restriction. The denominator cannot be zero, so x is not equal to -4.

Step 2: Multiply both sides by (x + 4) to clear the denominator.

$x = 2(x + 4)$

Step 3: Expand the right hand side.

$x = 2x + 8$

Step 4: Collect the x terms on one side.

$-x = 8$

Step 5: Divide by -1 to isolate x.

$x = -8.$

Check: x = -8 does not make the denominator zero.